

Experiment title: Developing a Full-Stack Web Application using React, Express.js, and MySQL for CRUD Operations.

Source Code: [GitHub Repository](#)

- 1) Launch Visual Studio Code and open the integrated terminal.
- 2) Create the main project directory, named experiment-9.
- 3) Navigate into the experiment-9 directory. Initialize the React application using **npx create-vite@latest**, naming the app frontend, and selecting the React framework with the JavaScript variant.
- 4) Navigate to the frontend/src directory and update the App.jsx component file.
- 5) Create a backend directory inside experiment-9. Inside backend, create the server.js file and populate it with the provided Node.js server code.
- 6) In the backend directory, initialize a Node.js project using **npm init -y** and install the required dependencies: **npm install express mysql2 cors**.
- 7) Set up the MySQL database connection. Configure the database credentials (host, user, password, database) in server.js. Execute the provided SQL commands to create the **testdb** database and the users table.
- 8) From the backend directory, start the server using **node server.js**.
- 9) Navigate to the frontend directory and launch the development server using **npm run dev**.

App.jsx

JavaScript

```
import { useEffect, useState } from "react";

function App() {
  const [users, setUsers] = useState([]);
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
```

```
const loadUsers = async () => {  
  const res = await fetch("http://localhost:3000/users");  
  const data = await res.json();  
  setUsers(data);  
};
```

```
const addUser = async () => {  
  await fetch("http://localhost:3000/users", {  
    method: "POST",  
    headers: {  
      "Content-Type": "application/json",  
    },  
    body: JSON.stringify({ name, email }),  
  });  
  setName("");  
  setEmail("");  
  loadUsers();  
};
```

```
useEffect(() => {  
  loadUsers();  
}, []);
```

```
return (  
  <div>  
    <h2>User Manager</h2>  
    <input  
      placeholder="Name"  
      value={name}  
      onChange={(e) => setName(e.target.value)}  
    />  
    <input  
      placeholder="Email"  
      value={email}  
      onChange={(e) => setEmail(e.target.value)}  
    />  
  </div>  
)
```

```

    />
    <button onClick={addUser}>Add User</button>
    <ul>
      {users.map((u) => (
        <li key={u.id}>
          {u.name} - {u.email}
        </li>
      ))}
    </ul>
  </div>
);
}

export default App;

```

Server.js

JavaScript

```

const express = require("express");
const cors = require("cors");
const mysql = require("mysql2");

const app = express();
app.use(cors());
app.use(express.json());

// 🛠️ MySQL Connection
const db = mysql.createConnection({
  host: "localhost",

```

```

    user: "root",
    password: "yourpassword", // change this
    database: "testdb",
  });

db.connect((err) => {
  if (err) {
    console.error("Database connection failed:", err);
    return;
  }
  console.log("MySQL Connected");
});

// GET users
app.get("/users", (req, res) => {
  db.query("SELECT * FROM users", (err, result) => {
    if (err) return res.status(500).send(err);
    res.json(result);
  });
});

// POST user
app.post("/users", (req, res) => {
  const { name, email } = req.body;

  if (!name || !email) {
    return res.status(400).send("Name and email required");
  }

  db.query(
    "INSERT INTO users (name, email) VALUES (?, ?)",
    [name, email],
    (err, result) => {
      if (err) return res.status(500).send(err);
      res.send("User added");
    }
  );
});

```

```
    );  
  });  
  
  const PORT = 3000;  
  app.listen(PORT, () => console.log(`Server running on port  
    ${PORT}`));
```

Database query :

```
SQL  
-- Create database  
CREATE DATABASE IF NOT EXISTS testdb;  
USE testdb;  
  
-- Create users table  
CREATE TABLE IF NOT EXISTS users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) NOT NULL  
);  
  
-- Insert seed data  
INSERT INTO users (name, email)  
VALUES ('user', 'user1@email.com');
```